WHERE A POSITIVE WEIGHTING ATTAINS ITS MAXIMUM THE MORPHISM IS FINITE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 22757-22788, immediately following the statement at 22748-22755.

Editorial note. One cross-chapter pointer (Descent) plus two same-chapter lemmas are recorded in assembly.external_references.

Editorial note. The displayed decomposition of the integral is retained verbatim; the conclusion W empty is drawn inside the span.

Allowed standard prerequisites (editorial).

- Upper semi-continuity of the integral. chapter lemma lemma-semicontinuous-int-w, cited at proof line 22758. Shows the locus where the integral equals its maximum is open.
- Etale splitting off of the quasi-finite part. chapter lemma lemma-etale-splits-off-quasi-finite-part-technical-variant, cited at proof lines 22760-22761. Provides the etale neighbourhood and the decomposition used to write the integral as a sum of a finite piece and the value at the point.
- Finiteness descends along the open etale morphism. Descent, lemma descent-lemma-descending-property-finite, cited at proof line 22783. Transfers finiteness from the etale neighbourhood back to f over the image of the open morphism.

Source reference. Original reference 02LA: focused-seeds/geometry/S9-morph-max-int-finite/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02LO: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 12136–12136. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F3J: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22695–22695. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a separated, locally quasi-finite morphism. Let $w : X \to \mathbf{Z}_{>0}$ be a positive weighting of f. Assume $\int_w f$ attains its maximum d and let $Y_d \subset Y$ be the open set of points y with $(\int_f w)(y) = d$. Then the morphism $f^{-1}(Y_d) \to Y_d$ is finite.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Observe that Y_d is open by Lemma 0F3J. Let $y \in Y_d$. Say $x_1, \ldots, x_n$ are the points of X lying over y. Apply Lemma 02LO to get an étale neighbourhood $(U, u) \to (Y, y)$ and a decomposition

$$U \times_Y X = W \amalg \coprod_{i=1,\ldots,n} \coprod_{j=1,\ldots,m_i} V_{i,j}$$

as in locus citatus. Observe that $d = \sum w(v_{i,j})$ where $w(v_{i,j}) = w(x_i)$. Since $\int_{V_{i,j} \to U} w|_{V_{i,j}}$ is locally constant by definition, we may after shrinking U assume these functions are constant with value $w(v_{i,j})$. We conclude that

$$\int_{U \times_Y X \to U} w|_{U \times_Y X} = \int_{W \to U} w|_W + \sum \int_{V_{i,j} \to U} w|_{V_{i,j}} = \int_{W \to U} w|_W + (\int_f w)(y)$$

This is $\geq (\int_f w)(y) = d$ and we conclude that W must be the emptyset. Thus $U \times_Y X \to U$ is finite. By Descent, Lemma 02LA this implies that $X \to Y$ is finite over the image of the open morphism $U \to Y$. In other words, we see that f is finite over an open neighbourhood of y as desired. $\square$